Quantitative Macro Theory **HW #7**
Spring 2019 **Due at 10pm on *Fri* 4/12/2019**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

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1. (60 points) **Equilibrium Firm Dynamics: Hopenhayn and Rogerson (1993)**

Calibration:

- Productivity follows AR(1) in logs

$$log(z') = (1-\rho)log(\bar{z}) + \rho log(z) + \sigma\epsilon'$$

  Approximate it wit a Markov chain with 20 values. Call the transition Γ. We use $\rho = 0.9$, $log(\bar{z}) = 1.39$ and $\sigma = 0.2$.
- Firms enter the industry with productivity z drawn from a uniform distribution, G, $\log(z) \in [1, 1.39]$.
- The number of firms entering will be initially set at $m = 0.1$. This will be solved in equilibrium later.
- Production function is zn^{α} where $\alpha = 2/3$.
- Since the period is 5 years, set $r = 0.20$.
- The cost of adjusting labor is $\delta \times max\{n - n', 0\}$, where $\delta = 0.1$ (6 months).
- The fixed cost firms pay to produce every period is $c_f = 20$.
- The fixed cost firms pay to enter the industry is $c_e = 40$.
- Finally, the total demand for the product is $D(p) = \theta/p$ with $\theta = 100$. Initially, we'll set $p = 0.90$. This will also be set in equilibrium.

(a) (10 points) The problem of a firm with productivity z and employment from the previous period n that decides to produce in this period.

$$V(z_i, n) = \max_{n'} p \times z_i n'^{\alpha} - n' - \delta \max\{n - n', 0\} - c_f + \frac{1}{1+r}\sum_{j=1}^{5}\Gamma_{i,j}\max\underbrace{\{V(z_j, n'), -\delta n'\}}_{\text{exit decision}}$$

Note that the choice of employment n' is dynamic because there is a cost of firing workers, δ per worker. In the next period, if the firm decides to shut down, it must pay $\delta n'$ because all workers would be fired.

As a solution to this Bellman equation, you will find to policy functions, $N(z,n)$ for employment and $X(z,n)$ for shutting down. You can solve it with any method we studied so far.[1]

Plot the functions V, N, X as a function of n for three values of z (low, middle, high). Explain your findings.

(b) (10 points) Compute the value of entry as

$$V^e(p) = -c_e + \frac{1}{1+r}\sum_{j=1}^{5} G_j \max\{V(z_j,0),0\}.$$

Find the value of $p = p^*$ such that $V^e = 0$. Show how it would vary for $c_e = 30$ and $c_e = 50$. Explain your findings.

To find the value of p such that $V^e(p) = 0$, write a function that takes as input p and solves V and then V^e. This function should return V^e. Call that function $F(p)$. Note that you need to find a zero for that function. For instance, you can find fsolve in Matlab or code bisection. A good initial guess for p is the value given before, 0.9.

(c) (10 points) Given the value of p solved above, we are going to solve for the distribution for firms. Let $\Lambda(z,n)$ be the number of firms with productivity z and previous employment n. Note that some of these firms will shut down in this period and other will produce employing $N(z,n)$. Thus, we are counting firms at the very beginning of the period (before they make their employment and exit decision).

Compute the stationary $\Lambda(z,n)$ and plot it in 3-D. [2]

To compute the stationary Λ you can iterate on the law of motion as we did for households. Guess a value of Λ_0 and given it compute Λ_1. For all j, l,

$$\Lambda_1(z_j,n_l) = \sum_{i,m} \Lambda_0(z_i,n_m)(1-X(z_i,n_m))\Gamma_{i,j} I_{N(z_i,n_m)=n_l} + mG(z_j)I_{n_l=0}$$

where I is an indicator function that it is 1 if the condition is satisfied and 0 otherwise.

[1] An option is to make a grid with 50 points for n from 0 to the the unconstrained optimal employment for the largest value of z in your grid.

[2] If you use Matlab, a nice way is using:
$[zz, nn] = meshgrid(z, n)$;
stem3(zz',nn',Λ,'Marker','s','MarkerEdgeColor','m','MarkerFaceColor','g');
The first line create a matrix for the grids z and n of the same dimension than Λ. The second line produces the plot.

(d) (10 points) Given the solution for Λ compute total output produced

$$Q = \sum_{i,m} \Lambda(z_i, n_m) z_i N(z_i, n_m)^{\alpha} (1 - X(z_i, n_m)).$$

Similarly, you can compute many statistics. For example, the total number of firms as

$$\sum_{i,m} \Lambda(z_i, n_m),$$

or the exit rate,

$$exit = \frac{\sum_{i,m} \Lambda(z_i, n_m) X(z_i, n_m)}{\sum_{i,m} \Lambda(z_i, n_m)},$$

or total labor employment by firms

$$L = \sum_{i,m} \Lambda(z_i, n_m) N(z_i, n_m)(1 - X(z_i, n_m))$$

Produce a table with all these statistics: entry rate, exit rate, total adjustment cost paid, total output, total employment, and output per worker.

(e) (10 points) Note that in equilibrium we want $Q = D(p) = \theta/p$. You solve for Q in part (c). If Q is not equal (or very close to) θ/p, adjust m and solve for Q again. Note that you don't need to solve for the value functions or p again. Recompute all the statistics in the previous questions. Did they change? Why?

2. (50 points) **Firing Costs**

(a) (25 points) Set the firing cost $\delta = \{0, 0.2\}$ and present the results of the equilibrium values of m^*, p^* together with the values of entry rate, exit rate, total adjustment cost paid, total output, total employment, aggregate productivity, and aggregate profits. Explain (economic intuition) what is the effect of increasing/eliminating firing costs.

(b) (25 points) Set the persistence $\rho = \{0.5, 0.75\}$ and present the results of the equilibrium values of m^*, p^* together with the values of entry rate, exit rate, total adjustment cost paid, total output, total employment, aggregate productivity, and aggregate profits. Explain (economic intuition) what is the persistence of productivity.